BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 061 · Debugging and Nested Constructs

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does sequence mean in programming?

Q2. What is 4 + 3 * 2?
A. 10
B. 14
C. 20
D. 24

Q3. What is 2 ^ 3 in OCR ERL?

Q4. Join text '12' to '3'. Then convert '12' to an integer and add 3. Give both results. What should happen if the input is 'twelve'?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Syntax error — An incorrect language keyword or missing ENDIF breaks the grammar.

Logic error — A running program may produce an incorrect result because its steps are wrong.

Nested selection — Each IF within an IF needs the correct closing ENDIF.

Debugging cycle — Test, locate, correct and retest the fault.

Algorithm — A sequence of precise problem-solving instructions.

Trace table — A table tracking variable values during execution.

New learning

Syntax error: An incorrect language keyword or missing ENDIF breaks the grammar. For example, consider a membership checker.

Logic error: A running program may produce an incorrect result because its steps are wrong. For example, consider a sports bands.

Nested selection: Each IF within an IF needs the correct closing ENDIF. For example, consider a ticket discounts.

Debugging cycle: Test, locate, correct and retest the fault. For example, consider a club rules.

Worked example

A membership checker is being developed. Explain how syntax error and logic error are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].

3. Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].

4. Give one accurate point and a linked reason for each technique.

We Do · Practise with guidance

Guided 1

A sports bands is being programmed. Explain why syntax error matters. [2 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].

Guided 2

For a ticket discounts, explain how logic error is used. [2 marks]

Worked method: Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].

Guided 3

A club rules needs nested selection and debugging cycle. Explain the roles of both. [4 marks]

Worked method: Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the logical operator so 260 is rejected.

```
octet = 260
if octet >= 0 OR octet <= 255 then
    print("Valid")
else
    print("Invalid")
endif
```

iterate

Question 1

State TWO concepts needed for this programming technique. [2 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 2

Explain how logic error and nested selection can help a ride height. [4 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 3

Explain the roles of syntax error, nested selection and debugging cycle when creating a reward scoring. [6 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

An incorrect language keyword or missing ENDIF breaks the grammar.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Instructions execute in the specified order.

Q2. A — 10

Q3. 8; ^ denotes exponentiation in OCR ERL.

Q4. Joining the strings gives '123'. Converting '12' to an integer and adding 3 gives 15. Handle or reject 'twelve': it cannot be cast to an integer directly.

I Do / We Do pages · Guided practice

Q1. Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].

Q2. Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].

Q3. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1].

You Do pages · Code First

```
octet = 260
if octet >= 0 AND octet <= 255 then
    print("Valid")
else
    print("Invalid")
endif
```

Expected result: Invalid

You Do pages · Independent questions

Q1. Syntax error [1]; Nested selection [1].

Q2. Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1]. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1]. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: